

Two errors, not one: the blade tones are bent by up to 8 dB below 1 kHz, and under them sits a broadband error of ± 1.3 dB (2.7 dB worst) that no band is exempt from. **The arc is not qualified to a standard anywhere yet.** For these runs the arc was laid flat (horizontal), with the propeller axis vertical. All microphones then sit at the same height in the propeller plane, so a perfect setup reads the same level everywhere: a circle. Twelve runs (prop7-18, 1-2 Sept 2026, PWM 1900); what is left after each microphone's own offset is removed is the room.

remedies report: docs/arc-validation-remedies.pdf



Taking the tones out removes the big errors, not all of them. A wide band averages a reflection out, a tone does not, which is why the tones carry the big errors. With every tone notched the polar is round to ± 1.3 dB in 92% of cells, worst 2.7 dB — but 99 of those 209 cells sit more than three standard errors from zero, so the rest is a real pattern, not scatter, and it is no smaller above 3 kHz than below (2.7 dB at +72°/2 kHz, 2.6 at -72°/397 Hz, 2.6 at +54°/5 kHz). The tones need the loudspeaker check or the speed ladder; the broadband floor of ± 1 -3 dB is there in every band (report \$00, \$03).

The propeller's mean field. With the arc flat and the axis vertical every microphone sees the propeller from the same angle, so an axisymmetric propeller cannot make one spot differ from another. Its inflow can, for the tones only; at 238 Hz the tones are bent and the broadband beside them is not, so this is not excluded (remedies report, §10).



1. **200–250 Hz, blue at -36° .** Something about 1.5 m across, off to one side of the rig, with a bunched path 0.7 m longer than the direct one at -36° : a wall, a table top or the corner within about half a metre of that microphone.
2. **630–1000 Hz, blue at -72° .** Something about 0.4 m across near the -90° end of the arc on the corner side: the stand base, a bar of the arc frame, a cable tray.
3. **3150 Hz, scattered red and blue.** Something about 0.1 m across at the microphones themselves: clamps, mounts, the cable loop.

+54° at 800 Hz: prop8 with 810-8901 reads +5.6 re gain = P +4.7 + M +1.4; prop18 with 811-1892 reads +3.1 = +4.7 - 4.2. Two mics, one room value.